

Sparse Attention Networks for Long Context Language Modeling

Abstract

We study attention complexity in Transformers and propose Sparse Attention on the WikiText-103 benchmark dataset.

Introduction

Standard dense attention requires quadratic memory. Sparse attention reduces computation while maintaining accuracy.

Method

Our method uses block-sparse attention masks across 16 attention heads on sequence lengths up to 8192 tokens.

Experiments

We train on the WikiText-103 dataset using 8 A100 GPUs with AdamW optimizer.

Results

Sparse attention mechanisms achieve superior performance and lower perplexity than dense attention on the WikiText-103 benchmark while reducing memory consumption by 80 percent.

Limitations

Hardware kernel support is required for optimal efficiency at sequence lengths below 512 tokens.

Conclusion

Sparse attention is strictly superior to dense attention for large scale language modeling.